

Project Proposals

for the

Deutsche Forschungsgemeinschaft

Hybrid Multiscale Modeling of Biological Aggregate Dynamics: Graph-Based Learning and Neural ODE Closures for Population Balance Models

(Keyword: Dynamics of Biological Aggregates in Bioreactors)

by

Prof. Dr.–Ing. Heiko Briesen

Dr.–Ing. Ali Khalifa

September 2026

Technical University of Munich

School of Life Sciences

Chair of Process Systems Engineering

Hybrid Multiscale Modeling of Biological Aggregate Dynamics: Graph-Based Learning and Neural ODE Closures for Population Balance Models

1 Starting point

1.1 State of the Art

Biological aggregates such as cell spheroids, organoids, microtissues, and other multicellular structures are increasingly important in bioprocess engineering. Their behavior is determined by interacting biological, transport, and mechanical processes. These include cell growth, nutrient and oxygen transport, cell–cell adhesion, aggregate restructuring, collisions, aggregation, and flow-induced loss of aggregate integrity.

The relevant mechanisms act across different spatial and temporal scales. At the primary-particle scale, the local coordination and connectivity of individual cells determine the internal structure and mechanical strength of an aggregate. At the aggregate scale, morphology influences hydrodynamic loading, mass transfer, and collision behavior. At the reactor scale, the combined evolution of many aggregates determines the population size distribution and, ultimately, bioprocess performance.

1.1.1 Population Balance Modeling of Aggregate Populations

Population balance models provide an established framework for describing the temporal and spatial evolution of distributed particle or aggregate populations. The population is represented by a number-density function defined over one or more internal coordinates. Depending on the application, these coordinates may include aggregate size, mass, age, composition, porosity, or biological state.

For aggregate-forming systems, the population balance equation commonly contains terms describing growth, aggregation, and deagglomeration. Aggregation is represented through a collision kernel, which is often decomposed into a collision frequency and a collision efficiency. Deagglomeration is described through a frequency function and a daughter-size distribution.

Multidimensional PBMs can, in principle, account for properties in addition to aggregate size. Models including morphology-related coordinates, such as porosity or fractal structure, have been proposed for different particulate systems. Nevertheless, the increased dimensionality substantially raises the numerical complexity of the population balance equation. Consequently, most practically applicable models retain aggregate size or mass as the dominant internal coordinate and represent structural effects through simplified correlations or prescribed auxiliary variables.

For biological aggregates, PBMs have been applied to predict the evolution of aggregate-size distributions resulting from growth and aggregation. Such models demonstrate the suitability of the PBM framework for biological populations. However, the corresponding kernels generally remain system-specific and must be identified from experimental population data or adopted from simplified collision and aggregation theories.

1.1.2 Closures for Aggregation and Deagglomeration

The predictive quality of a PBM depends strongly on the closure models used for aggregation and deagglomeration. Collision-frequency expressions are commonly derived from idealized mechanisms such as Brownian motion, differential settling, laminar shear, or turbulent fluctuations. The probability that a collision produces a stable aggregate is introduced through a collision-efficiency factor.

These closures generally depend on aggregate size, fluid properties, local shear rate, or turbulent energy-dissipation rate. Aggregate morphology may be included through quantities such as fractal dimension, porosity, collision radius, or an effective density. However, these quantities are commonly prescribed or obtained from separate empirical relationships.

A similar limitation applies to deagglomeration models. Deagglomeration frequencies are usually related to the comparison between an external hydrodynamic stress and an assumed aggregate strength. Daughter-size distributions are then prescribed using empirical or idealized distribution functions.

Such models provide computationally tractable closures, but they reduce the detailed mechanical state of an aggregate to a small number of averaged quantities. They do not directly resolve how hydrodynamic loading is transmitted through the aggregate contact network, where mechanically weak regions develop, or how previous loading changes the subsequent response of an aggregate.

1.1.3 Coupling of CFD and Population Balance Models

Coupling PBMs with computational fluid dynamics allows spatially varying reactor conditions to be considered. CFD can provide local quantities such as velocity, shear rate, turbulent energy-dissipation rate, and scalar concentrations. These quantities are then used as inputs to local aggregation and deagglomeration kernels.

CFD–PBM coupling therefore improves upon models based exclusively on spatially averaged reactor conditions. It allows aggregate populations passing through different reactor regions to experience different collision frequencies and mechanical environments.

However, CFD–PBM coupling does not by itself resolve the closure problem. The local flow quantities obtained from CFD must still be translated into aggregation probabilities, structural changes, and deagglomeration frequencies through prescribed kernel expressions. The aggregates are generally not resolved geometrically and their internal contact structures are not represented.

1.1.4 Particle-Resolved Modeling of Aggregates

Discrete element methods represent an aggregate as an assembly of primary particles connected through contact or adhesive interactions. They can provide direct information about particle positions, coordination, contact forces, bond loading, restructuring, and bond rupture.

When DEM is coupled with a fully resolved fluid model, the fluid motion around the individual primary particles can additionally be calculated. Particle-resolved CFD–DEM can therefore quantify hydrodynamic force distributions, shielding effects, local flow gradients, scalar transport, and the mechanical loading transmitted through the aggregate.

These simulations provide information that is inaccessible to conventional CFD–PBM models. In particular, they allow aggregates with equal size but different morphology and connectivity to be subjected to identical external conditions and compared directly.

Because of their high computational cost, however, particle-resolved simulations are restricted to individual aggregates, binary collisions, or small aggregate ensembles under canonical flow conditions. They cannot be applied directly to complete reactor-scale aggregate populations. Their main value for population modeling therefore lies in providing mechanistic reference data for reduced closure models.

1.1.5 Reduced and Data-Driven Representations of Aggregate Structure

Conventional scalar morphology descriptors, including porosity, fractal dimension, radius of gyration, coordination number, and shape factors, summarize selected aspects of aggregate structure. They are useful for comparing aggregate classes but may not uniquely describe the internal connectivity of an aggregate.

Two aggregates may possess similar size, porosity, and fractal dimension while differing in the spatial distribution of contacts, load-bearing pathways, weakly connected branches, and exposed surface regions. Such differences may substantially affect their response to hydrodynamic loading and collisions.

Graph-based representations provide a natural description of aggregates because primary particles can be represented as nodes and contacts or adhesive bonds as edges. Graph neural networks can

process aggregates containing different numbers of primary particles and can combine geometric, mechanical, hydrodynamic, and transport-related features.

In this context, a graph-based reduced state could retain structural information that is lost when an aggregate is represented only through its size and a limited set of averaged morphology descriptors. However, the suitability of such representations for developing population-balance closures for biological aggregates remains to be established.

1.1.6 Continuous-Time Models and Hydrodynamic History

Most PBM closures are memoryless. At a given time, the aggregation or deagglomeration rate is calculated from the current particle properties and the instantaneous local flow conditions.

This assumption may be insufficient when aggregates undergo gradual restructuring, weakening, or cumulative bond damage. Two aggregates with the same current size and similar instantaneous morphology may respond differently if they have previously experienced different hydrodynamic loading histories.

Continuous-time data-driven models, including Neural ODEs, provide a possible framework for introducing evolving internal state variables. Such variables may represent structural adaptation or accumulated mechanical damage and can be updated as an aggregate moves through changing flow conditions.

The relevant question is not whether Neural ODEs can replace the physical models, but whether the inclusion of a learned internal state improves the prediction of aggregate evolution compared with instantaneous, memoryless closures.

1.1.7 Scope of Biological Growth and Viability

Biological growth, metabolic activity, and viability are important for the long-term evolution of cell aggregates. Their prediction requires biological kinetic information relating nutrient availability and other environmental conditions to cellular responses.

Particle-resolved CFD–DEM can provide local transport quantities, such as oxygen or substrate concentrations and fluxes, but it does not directly generate biological growth, division, or death data. Such simulations alone therefore cannot be used to identify growth or viability kinetics.

In the proposed project, biological growth and viability will not be learned from CFD–DEM simulations. If required for the final reactor-scale application, they will be included through established biological kinetic models or experimentally determined correlations. The particle-resolved learning framework will focus on aggregate morphology, hydrodynamic loading, restructuring, aggregation, mechanical damage, and deagglomeration.

1.1.8 Research Gap

The main limitation of current aggregate PBMs is not that they are mathematically restricted to particle size. Rather, the central problem is the absence of transferable and mechanistically informed closures that connect aggregate structure and hydrodynamic history to aggregation, restructuring, and deagglomeration.

Existing approaches present several related limitations:

1. Aggregate morphology is commonly represented through size alone or through a small number of prescribed scalar descriptors. The internal contact topology and spatial distribution of mechanically weak regions are not retained.
2. CFD–PBM approaches provide spatially resolved environmental conditions but still require empirical or semi-empirical expressions to translate these conditions into aggregation and deagglomeration rates.
3. Particle-resolved simulations can reveal morphology-dependent hydrodynamic and mechanical

behavior, but no systematic method is available for transferring this high-dimensional information into computationally efficient PBM closures.

4. Conventional closure models generally depend on instantaneous conditions and therefore cannot represent persistent restructuring or cumulative mechanical damage caused by previous hydrodynamic exposure.
5. Kernels calibrated for one aggregate class or operating condition often have limited transferability to different morphologies, cohesion strengths, or reactor conditions.

The opportunity for improvement therefore lies in developing reduced aggregate states that preserve the particle-scale structural information relevant to population dynamics and in describing how these states evolve under time-dependent hydrodynamic loading.

The proposed hybrid framework addresses this need by using particle-resolved CFD–DEM to generate mechanistic reference data, graph-based models to identify structure-sensitive aggregate representations, and continuous-time models to describe restructuring and damage accumulation. The resulting reduced models will be used to formulate morphology-dependent and history-dependent aggregation and deagglomeration closures for reactor-scale PBMs.

1.2 Preliminary Work

2 Objectives and Work Programme

2.1 Objectives

The overall objective of the proposed project is to develop a hybrid multiscale modeling framework that transfers information from particle-resolved simulations of biological aggregates to reactor-scale population balance models.

The project focuses on aggregate morphology, hydrodynamic loading, transport conditions, mechanical integrity, aggregation, restructuring, and deagglomeration. Biological growth and viability are treated as external biological processes and are not inferred from CFD–DEM data.

The following four research hypotheses guide the project:

1. **H1:** Aggregate size alone is insufficient to predict hydrodynamic loading, transport behavior, restructuring, and deagglomeration. Graph-based representations of aggregate morphology provide additional predictive information beyond conventional scalar descriptors.
2. **H2:** The response of an aggregate depends not only on its instantaneous structure and environmental conditions but also on its previous hydrodynamic and mechanical exposure. A history-aware Neural ODE can therefore predict aggregate evolution more accurately than a memoryless closure model.
3. **H3:** Aggregation and deagglomeration closures can be learned from particle-resolved simulations in a transferable form that generalizes across aggregate sizes, morphologies, cohesion strengths, and hydrodynamic conditions not explicitly contained in the training data.
4. **H4:** State-aware PBMs using graph-based and history-dependent closures provide more accurate reactor-scale predictions of aggregate population dynamics than conventional PBMs based only on aggregate size and instantaneous bulk conditions.

The four hypotheses are addressed through four interconnected work packages. Each work package produces a defined output that forms the basis for the next stage of the project.

2.2 Work Programme Including Proposed Research Methods

2.2.1 WP1: Generation of Particle-Resolved and DEM-Based Aggregate Data

Objective and general strategy The objective of WP1 is to generate a structured database describing the hydrodynamic loading, structural evolution, collision response, aggregation, and deag-

glomeration of model biological aggregates. The work package will combine high-fidelity particle-resolved direct numerical simulations with computationally less expensive DEM-only simulations.

The two simulation levels serve different purposes. Particle-resolved CFD–DEM will be used to determine how the surrounding liquid and the instantaneous aggregate morphology affect the distribution of hydrodynamic loading within an aggregate. DEM-only simulations will be used for larger parameter studies in which the contact-mechanical and collision parameters must be varied systematically. Selected cases will be simulated with both approaches in order to determine when the simplified DEM-only treatment is sufficient and when fluid-mediated effects must be retained.

The complete procedure consists of four steps:

1. generation and mechanical calibration of bonded model aggregates;
2. particle-resolved DNS–CFD–DEM simulations of single aggregates under controlled turbulent and canonical flow conditions;
3. DEM-only simulations of binary aggregate collisions and mechanically prescribed loading histories;
4. processing of the resulting particle, bond, fluid, collision, and deagglomeration data into graph-based trajectories and aggregate-level descriptors.

WP1.1: Definition and generation of model aggregates Individual biological cells will be represented by spherical primary particles. The initial reference system will use monodisperse particles to separate the influence of aggregate morphology from that of primary-particle size distribution. Polydispersity may subsequently be introduced as an additional variation if it is relevant for the selected biological system.

The physical primary-particle diameter, density, and mechanical properties will not be selected as arbitrary DEM parameters. They will be defined for a specific reference cell or aggregate system using experimental data and values reported in the biological and biomechanical literature. The following quantities will be required:

- representative primary-cell diameter and its variation;
- primary-particle density relative to the surrounding liquid;
- effective contact stiffness and damping;
- interparticle friction or tangential resistance;
- adhesive bond strength under normal, tangential, bending, and torsional loading;
- characteristic aggregate porosity, coordination number, and macroscopic strength.

Because cellular contacts and extracellular-matrix connections are not adequately represented by a purely distance-dependent van der Waals force, aggregate integrity will be modeled using finite-strength adhesive bonds. A parallel-bond-type model will be employed in which an adhesive connection can transfer normal and tangential forces as well as bending and torsional moments. Bond rupture occurs when a prescribed stress- or force-based failure criterion is exceeded.

The microscopic bond parameters will be calibrated against aggregate-scale mechanical targets rather than assigned independently. Depending on the availability of experimental data, the calibration targets may include aggregate compression response, effective elastic modulus, tensile strength, critical shear stress, or force required to separate two connected cells. If no complete experimental dataset is available, a physically plausible parameter range will be defined and propagated through the simulations as an explicit uncertainty range.

Several controlled aggregate classes will be generated. The initial design will include aggregates consisting of approximately

$$N_p \in 50, 100, 250, 500 \quad (1)$$

primary particles. A limited number of larger aggregates containing approximately 1000 primary particles will be considered for scale-generalization tests if computational resources permit.

For each particle number, several structural classes will be created:

- compact approximately spherical aggregates;
- porous aggregates with controlled void fraction;
- fractal or diffusion-limited structures with controlled fractal dimension;
- anisotropic or elongated aggregates;
- aggregates with intentionally introduced weakly connected regions.

The structural variation will allow aggregates with similar mass or equivalent diameter but different contact topology to be compared directly.

Each generated aggregate will first undergo a relaxation and stabilization procedure in DEM. The aggregate will be considered mechanically equilibrated when its total kinetic energy, maximum particle velocity, and unbalanced contact force remain below prescribed tolerances for a defined relaxation interval. The resulting particle positions, contact network, bond forces, porosity, coordination number, radius of gyration, and principal dimensions will be stored as the initial aggregate state.

WP1.2: Particle-resolved DNS–CFD–DEM simulations The high-fidelity simulations will be performed using resolved CFDEM coupling based on a fictitious-domain or immersed-boundary treatment. The fluid motion around the individual primary particles will be spatially resolved. No unresolved drag correlation will therefore be used as the primary description of the fluid–particle interaction.

The fluid equations will be solved by direct numerical simulation. This means that the instantaneous flow field and all dynamically relevant turbulent scales represented in the computational domain will be resolved without applying a RANS turbulence model. The designation DNS refers here to the fluid-phase treatment, while the moving primary particles and their interactions are resolved through the coupled DEM model.

The principal turbulent setup will consist of a triply periodic cubic domain containing one initially stabilized aggregate. Homogeneous isotropic turbulence will be maintained through large-scale forcing. The periodic box provides a controlled environment in which turbulence intensity, dissipation rate, Kolmogorov scales, and aggregate-to-flow scale ratios can be varied independently of reactor geometry.

The simulation procedure will consist of the following stages:

1. A single-phase turbulent flow will first be generated in the periodic domain.
2. The flow will be continued until statistically stationary turbulence is obtained. Stationarity will be assessed from the time evolution of the domain-averaged kinetic energy, dissipation rate, and selected velocity statistics.
3. A mechanically equilibrated aggregate will then be inserted into the statistically stationary flow field.
4. The coupled DNS–CFD–DEM simulation will be continued while resolving particle motion, fluid–particle momentum transfer, contact loading, bond deformation, restructuring, and possible bond rupture.

The box size and numerical resolution will be selected through preliminary grid and domain-size studies. Each primary particle must be represented by a sufficient number of CFD cells across its diameter to obtain converged hydrodynamic forces and local flow gradients. The periodic domain must also be sufficiently larger than the aggregate to minimize direct interaction with its periodic images.

The primary control parameters will include:

- aggregate particle number and equivalent diameter;
- aggregate morphology, porosity, and anisotropy;
- bond strength and bond-stiffness parameters;
- turbulent Reynolds number;
- mean turbulent energy-dissipation rate;
- ratio between aggregate size and Kolmogorov length scale;

- primary-particle and aggregate Stokes numbers;
- fluid viscosity and density;
- optional scalar diffusivity and surface-consumption boundary conditions.

A full factorial variation of all parameters would be computationally infeasible. The high-fidelity database will therefore be constructed using a staged design. First, a baseline aggregate and flow case will be used for verification, grid convergence, and time-step sensitivity. Second, individual physical groups will be varied systematically. Finally, a space-filling design will be used for the combined parameter study.

In addition to homogeneous isotropic turbulence, selected canonical flows may be included to isolate individual loading mechanisms. These may comprise simple shear flow, planar extensional flow, and controlled oscillatory loading. Such cases provide interpretable reference conditions for determining how aggregate response changes between approximately steady and strongly fluctuating hydrodynamic forcing.

Stopping criteria for the particle-resolved simulations A simulation will not be stopped merely after an arbitrary fixed physical time. Different stopping criteria will be applied depending on the observed aggregate response.

For non-deagglomerating aggregates, the simulation will continue until the aggregate has experienced a prescribed number of large-eddy turnover times and its structural descriptors have either reached a statistically stationary regime or shown no systematic change over a defined observation window.

For restructuring cases, the simulation will continue until the principal structural quantities, including porosity, coordination number, radius of gyration, and contact topology, approach a new statistically stable state.

For deagglomerating cases, the simulation will continue beyond the first bond rupture. It will be stopped only after the resulting connected components are clearly separated and their particle membership no longer changes over a prescribed interval. This is necessary to distinguish a local bond rupture from an actual deagglomeration event.

A deagglomeration event will be defined using the connectivity graph of the aggregate. The original aggregate is considered separated when its bond graph splits into two or more persistent connected components. The daughter components must remain disconnected and move apart for a minimum confirmation interval.

Additional safety stopping criteria will include excessive particle displacement, loss of numerical stability, violation of the target Courant number, or completion of a predefined maximum exposure time.

Data sampled from particle-resolved simulations The stored data will be divided into fluid-level, particle-level, bond-level, and aggregate-level quantities.

At the fluid level, the following quantities will be sampled:

- instantaneous velocity and pressure fields;
- local velocity-gradient and strain-rate tensors;
- turbulent kinetic energy and dissipation-related descriptors;
- immersed-boundary or fictitious-domain forcing field;
- optional oxygen or substrate concentration fields and surface fluxes.

At the primary-particle level, the following quantities will be stored:

- particle position, velocity, and angular velocity;
- hydrodynamic force and torque;
- particle acceleration;
- local fluid velocity and local velocity-gradient descriptors;
- local scalar concentration and flux, if scalar transport is included;

- particle coordination number and surface-exposure indicators.

At the bond level, the following quantities will be sampled:

- identities of the two connected particles;
- normal and tangential bond forces;
- bending and torsional moments;
- bond deformation and normalized loading;
- bond age;
- time and mode of bond rupture.

At the aggregate level, the following descriptors will be calculated:

- total particle number and number of connected components;
- equivalent diameter and radius of gyration;
- porosity or compactness;
- coordination-number distribution;
- principal dimensions and anisotropy;
- fractal dimension, where applicable;
- total hydrodynamic force and torque;
- distributions and extrema of particle and bond loading;
- number and spatial distribution of broken bonds;
- daughter sizes and daughter morphologies after deagglomeration.

The complete fields will be written only at selected times because storing every CFD field at every coupling step would be prohibitively expensive. Particle and bond quantities will be sampled at a higher temporal frequency than the complete three-dimensional flow fields. Event-triggered output will additionally be used around bond-rupture, restructuring, collision, and deagglomeration events.

WP1.3: DEM-only simulations DEM-only simulations will be used to explore parameter combinations that cannot be covered economically using particle-resolved CFD–DEM. Their purpose is not to replace the high-fidelity liquid-phase simulations, but to isolate contact-mechanical mechanisms and generate statistically meaningful collision-outcome data.

The main DEM-only application will be the simulation of binary collisions between two initially stabilized aggregates. A typical setup will consist of two aggregates positioned at a prescribed initial separation. Translational and, where relevant, rotational velocities will be assigned to produce a specified relative velocity and collision geometry.

The principal collision parameters will include:

- particle numbers and size ratio of the two aggregates;
- morphology and orientation of both aggregates;
- relative collision velocity;
- impact parameter or collision angle;
- initial angular velocities;
- normal, tangential, bending, and torsional bond strengths;
- contact stiffness, damping, and friction;
- structural heterogeneity or intentionally weakened regions.

The collision simulations will classify the final outcome as:

- rebound without persistent structural change;
- rebound after restructuring or erosion;
- temporary contact followed by separation;
- permanent sticking;

- sticking accompanied by restructuring;
- collision-induced deagglomeration of one or both aggregates.

For each parameter combination, repeated simulations with different orientations and aggregate realizations will be performed. This provides a probability distribution of collision outcomes rather than a single deterministic label.

The DEM-only simulations will also be used for controlled mechanical loading experiments. An aggregate may be subjected to prescribed force, torque, shear-like differential forcing, cyclic loading, or sequences of loading and relaxation. These simulations will establish whether the aggregate response depends on its loading history even when the instantaneous loading is identical.

The absence of an explicitly resolved fluid in these simulations substantially reduces the computational cost and allows systematic variation of collision and bond parameters. However, DEM-only results cannot automatically be interpreted as collisions in vacuum or transferred directly to liquid systems. The initial velocities and damping parameters will be selected to represent collision conditions obtained from reactor-scale flow estimates or sampled from the particle-resolved simulations.

Selected DEM collision cases will therefore be repeated with fully resolved CFD–DEM. The comparison will quantify changes caused by liquid-mediated viscous dissipation, hydrodynamic shielding, and fluid forces during approach and separation. From this comparison, a domain of validity for the DEM-only collision model will be established.

Stopping criteria for DEM-only collision simulations A binary collision simulation will be considered complete when one of the following persistent outcomes has been reached:

1. the aggregates have separated beyond a prescribed distance and their relative motion indicates continued separation;
2. a stable combined aggregate has formed, its contact topology remains unchanged, and its kinetic energy has fallen below a prescribed tolerance;
3. one or both aggregates have separated into persistent connected components whose membership remains unchanged over a confirmation interval;
4. a predefined maximum simulation time has been reached without a clear classification.

Cases reaching the maximum time without a stable classification will be labeled as unresolved or long-duration contacts and analyzed separately rather than forced into a sticking or rebound class.

WP1.4: Data processing and preparation for machine learning The raw simulation data will be transformed into synchronized aggregate trajectories. At each sampled time, the aggregate contact network will be converted into a graph in which primary particles form the nodes and active adhesive bonds form the edges.

Node features will be assembled from particle position, motion, hydrodynamic loading, local flow descriptors, coordination, and optional scalar-transport quantities. Edge features will be assembled from bond forces, moments, deformation, age, strength, and rupture state.

The raw quantities will be converted into dimensionless or normalized form wherever possible. Relevant dimensionless groups will include Reynolds, Stokes, cohesion, and aggregate-to-turbulence scale ratios. This step is required to improve transferability across particle sizes, fluid properties, and operating conditions.

Each trajectory will be assigned labels and targets including:

- time-dependent aggregate morphology;
- restructuring rate;
- accumulated and localized mechanical loading;
- time to first bond rupture;
- time to persistent deagglomeration;
- deagglomeration frequency or event probability;
- daughter-particle-number and daughter-size distributions;

- binary collision outcome;
- sticking probability under specified collision conditions;
- post-collision aggregate morphology.

Training, validation, and test datasets will be separated by complete aggregate realizations and operating conditions rather than by randomly dividing individual time steps. In addition, selected particle numbers, morphologies, bond strengths, and flow conditions will be withheld entirely from training to evaluate generalization.

The final outputs of WP1 are:

1. a calibrated bonded-particle representation of model biological aggregates;
2. a particle-resolved DNS–CFD–DEM database for single-aggregate loading, restructuring, and deagglomeration;
3. a DEM-only database of binary collision outcomes and controlled loading histories;
4. a cross-validation dataset comparing DEM-only and fluid-resolved collision simulations;
5. graph-structured, time-resolved datasets for the development of the GNN and Neural ODE models in the subsequent work packages.

2.2.2 WP2: Graph-Based Representation of Aggregate Structure

The objective of WP2 is to develop a reduced representation of biological aggregates from the particle-resolved data generated in WP1.

Each aggregate will be represented as a graph:

$$\mathcal{G}_a(t) = (\mathcal{V}_a(t), \mathcal{E}_a(t)), \quad (2)$$

where the nodes $\mathcal{V}_a$ represent primary particles and the edges $\mathcal{E}_a$ represent contacts or adhesive bonds.

Node features will contain particle-level information including relative particle position, local coordination, oxygen and substrate concentrations, hydrodynamic forces, particle motion, and local flow descriptors.

Edge features will contain mechanical information including normal and tangential bond forces, bond moments, bond age, and remaining bond strength.

A graph neural network encoder will map the particle-resolved aggregate graph onto a reduced representation:

$$\mathbf{z}_a = \mathcal{E}_\theta(\mathcal{G}_a), \quad (3)$$

where $\mathbf{z}_a$ represents a low-dimensional aggregate state.

The graph representation will be trained to retain information relevant to aggregate morphology, internal transport limitations, mechanical integrity, restructuring, and deagglomeration.

Auxiliary prediction tasks will be introduced to improve the physical interpretability of the learned representation. These may include the prediction of aggregate porosity, effective mass-transfer properties, mechanical damage indicators, and deagglomeration propensity.

The GNN-based representation will be compared with simpler models using only aggregate size and conventional morphology descriptors. This comparison provides the direct test of Hypothesis H1.

The main output of WP2 is a validated graph-based reduced representation that maps aggregates with variable numbers of primary particles onto a common low-dimensional state space.

2.2.3 WP3: History-Aware Neural ODE Closures for Aggregate Evolution

The objective of WP3 is to develop continuous-time reduced models describing aggregate restructuring and mechanical damage under changing environmental conditions.

The interpretable aggregate state is defined as:

$$\mathbf{x}_a = \begin{bmatrix} \phi_a \\ D_a \end{bmatrix}, \quad (4)$$

where ϕ_a denotes an aggregate structural variable such as porosity or compactness and D_a denotes an accumulated mechanical-damage variable.

The temporal evolution of the aggregate state will be described by a hybrid Neural ODE:

$$\frac{d\mathbf{x}_a}{dt} = \mathbf{f}_\theta(\mathbf{x}_a, \mathbf{z}_a, \mathbf{u}_a), \quad (5)$$

where $\mathbf{z}_a$ is the graph-based morphology representation from WP2 and $\mathbf{u}_a$ contains environmental and mechanical information obtained from CFD–DEM.

The system is written component-wise as:

$$\begin{aligned} \frac{d\phi_a}{dt} &= f_{\phi,\theta}(\phi_a, D_a, \mathbf{z}_a, \mathbf{u}_a), \\ \frac{dD_a}{dt} &= f_{D,\theta}(\phi_a, D_a, \mathbf{z}_a, \mathbf{u}_a). \end{aligned} \quad (6)$$

The environmental input vector may be defined as:

$$\mathbf{u}_a = \begin{bmatrix} \varepsilon_a \\ \dot{\gamma}_a \\ S_a \\ B_a \end{bmatrix}, \quad (7)$$

where ε_a is the local turbulent energy-dissipation rate, $\dot{\gamma}_a$ is a strain-rate descriptor, S_a contains hydrodynamic-loading statistics, and B_a contains bond-state and bond-rupture statistics.

The Neural ODE will learn how repeated hydrodynamic loading and internal bond stresses produce morphological restructuring and accumulate into a changing damage state.

To test Hypothesis H2, the history-aware model will be compared with memoryless models of the form:

$$S_{b,a} = g_{b,\theta}(\mathbf{z}_a, \mathbf{u}_a), \quad (8)$$

which use only the instantaneous state and operating conditions.

A non-negative deagglomeration frequency will be predicted from the evolving morphology and damage states:

$$S_{b,a} = g_{b,\theta}(\phi_a, D_a, \mathbf{z}_a, \mathbf{u}_a). \quad (9)$$

The probability of a deagglomeration event within a time interval Δt is then given by:

$$P_{\text{deagg},a}(\Delta t) = 1 - \exp(-S_{b,a}\Delta t). \quad (10)$$

The corresponding daughter-size distribution will be learned from deagglomeration events observed in the particle-resolved simulations.

Aggregation will be treated using a pairwise model. The graph representations of two colliding aggregates, together with relative velocity, collision geometry, aggregate size, and mechanical properties, will be used to predict the probability of permanent aggregation:

$$P_{\text{stick},ij} = g_{\text{agg},\theta}(\mathbf{z}_i, \mathbf{z}_j, \mathbf{q}_{ij}), \quad (11)$$

where $\mathbf{q}_{ij}$ contains the collision conditions.

The resulting aggregation kernel will be expressed as:

$$K_{ij}^{\text{agg}} = \beta_{ij}^{\text{coll}} P_{\text{stick},ij}, \quad (12)$$

where β_{ij}^{coll} denotes the collision frequency obtained from the reactor-scale flow and population model.

WP3 tests Hypothesis H2 through the comparison between history-aware and memoryless closures. It also tests Hypothesis H3 through validation on unseen aggregate morphologies, cohesion strengths, and hydrodynamic conditions.

The main outputs of WP3 are a history-aware Neural ODE for morphology and damage evolution, a deagglomeration-frequency closure, a daughter-size distribution, and a pairwise collision-efficiency model.

2.2.4 WP4: Integration into State-Aware Population Balance Models

The objective of WP4 is to integrate the developed closure models into a population balance framework applicable to reactor-scale simulations.

The PBM will describe aggregate populations using conventional internal coordinates such as aggregate size, augmented by reduced aggregate-state information where required.

The graph-based and Neural ODE models will provide closures for aggregation, restructuring, and deagglomeration without requiring particle-resolved simulations at reactor scale.

Growth and viability will not be predicted from CFD–DEM or learned within the particle-resolved framework. If they are required for a specific bioprocess application, they will be incorporated as external biological source terms based on established kinetic models or experimental correlations.

The resulting PBM framework will be coupled to coarse CFD simulations providing spatially and temporally varying hydrodynamic and transport conditions.

The model will predict aggregate-size distributions, structural-state distributions, aggregation dynamics, deagglomeration rates, and the response of aggregate populations to changing reactor operating conditions.

The state-aware PBM will be compared against conventional PBMs using size-dependent and memoryless closures. Validation will be performed using unseen particle-resolved simulations, dilute multi-aggregate simulations, and available experimental population data.

WP4 tests Hypothesis H4 by determining whether the graph-based and history-dependent closures improve population-level predictions. It also provides the final system-level test of Hypothesis H3.

The main output of WP4 is a reactor-scale hybrid multiscale simulation framework linking particle-resolved aggregate physics to population-level bioprocess predictions.

Table 1 summarizes the entire work programme.

Work packages	first year	second year	third year	PM
Objective A: Generation of breakup database				
A 1: Generation of agglomerates with specific morphologies	■			2
A 2: Extensive DNS-DEM simulations for individual agglomerates	■■■■■			12
Objective B: Analyzing the deagglomeration behavior for deeper physical insight				
B 1: In-depth physical analysis		■■■■■		4
B 2: Validation of existing theory-based models		■		2
Objective C: Developing a novel model for the fluid-induced agglomerate breakup using machine learning and incorporating it into the E–L framework				
C 1: Assessment of different machine-learning techniques (Symbolic regression based on sparse regression or genetic programming)		■■■■■		4
C 2: Designing and training the novel model		■■■■■		6
C 3: Incorporation of the novel model into the E–L framework			■	1
C 4: Study of practically relevant test cases			■■■■■	4
Documentation and publication of the results including storage of main data in NFDI	■	■	■	1

Table 1: Time schedule of the work packages within the project (PM = person-months).

2.3 Handling of research data

As done for previous test cases related to FSI studies, all data which might be interesting for the entire community will be made available online including a detailed documentation and best practice advices. For this purpose, a membership application will be submitted to the German National Research Data Infrastructure for Engineering Sciences (NFDI4Ing) headed by RWTH Aachen to securely store the primary data as well as the source codes and scripts for their analysis. These data will be prepared in compliance with the established recommendations of this organization to assure a proper research data management (RDM) that implements the FAIR data principles: Data has to be findable, accessible, interoperable, and re-usable. With regard to the seven derived archetypes being representative for the majority of the engineering sciences, DORIS (high performance measurement & computation) fits best to the current project.

2.4 Relevance of sex, gender and/or diversity

As is customary with job postings at HSU Hamburg, we will specifically look for female applicants and, if appropriate, invite them to apply. Increasing the proportion of women, particularly in technical/scientific subjects, is a high priority at HSU.

3 Project- and subject-related list of publications

Ten own publications that are most relevant to the project are highlighted in red.

- [1] G. Akiki and S. Balachandar. Shear-induced lift force on spheres in a viscous linear shear flow at finite volume fractions. *Phys. Fluids* 32 (11) (2020), 113306.
- [2] M. Alletto and M. Breuer. One-way, two-way and four-way coupled LES predictions of a particle-laden turbulent flow at high mass loading downstream of a confined bluff body. *Int. J. Multiphase Flow* 45 (2012), 70–90.
- [3] N. Almohammed and M. Breuer. Towards a deterministic composite collision outcome model for surface-tension dominated droplets. *Int. J. Multiphase Flow* 110 (2019), 1–17.
- [4] Y. Ammar, A. Dehbi and M. W. Reeks. Break-up of aerosol agglomerates in highly turbulent gas flow. *Flow Turbulence Combustion* 89 (3) (2012), 465–489.
- [5] M. U. Bäbler, L. Biferale, L. Brandt, U. Feudel, K. Guseva, A. S. Lanotte, C. Marchioli, F. Picano, G. Sardina, A. Soldati and F. Toschi. Numerical simulations of aggregate breakup in bounded and unbounded turbulent flows. *J. Fluid Mech.* 766 (2015), 104–128.
- [6] M. U. Bäbler, L. Biferale and A. S. Lanotte. Breakup of small aggregates driven by turbulent hydrodynamic stress. *Physical Review E* 85 (2) (2012), 025301.
- [7] M. U. Bäbler, M. Morbidelli and J. Baldyga. Modelling the breakup of solid aggregates in turbulent flows. *J. Fluid Mech.* 612 (2008), 261–289.
- [8] P. Bagchi and S. Balachandar. Shear versus vortex-induced lift force on a rigid sphere at moderate Re. *J. Fluid Mech.* 473 (2002), 379–388.
- [9] P. Bagchi and S. Balachandar. Effect of turbulence on the drag and lift of a particle. *Phys. Fluids* 15 (11) (2003), 3496–3513.
- [10] S. Balachandar. A scaling analysis for point-particle approaches to turbulent multiphase flows. *Int. J. Multiphase Flow* 35 (9) (2009), 801–810.
- [11] S. Balachandar, W. Moore, G. Akiki and K. Liu. Towards particle-resolved accuracy in Euler-Lagrange simulations of multiphase flow using machine learning and pairwise interaction extended point-particle (PIEP) approximation. *Theor. Comput. Fluid Dyn.* 34 (2020), 401–428.
- [12] A. Beck, D. Flad and C.-D. Munz. Deep neural networks for data-driven LES closure models. *J. Comput. Phys.* 398 (2019), 108910.
- [13] V. Becker, E. Schlauch, M. Behr and H. Briesen. Restructuring of colloidal aggregates in shear flows and limitations of the free-draining approximation. *J. Colloid Interface Science* 339 (2) (2009), 362–372.
- [14] S. Beetham and J. Capecehatro. Multiphase turbulence modeling using sparse regression and gene expression programming. *Nuclear Technology* 209 (12) (2023), 1977–1986.
- [15] S. Beetham and J. Capecehatro. Formulating turbulence closures using sparse regression with embedded form invariance. *Phys. Rev. Fluids* 5 (8) (2020), 084611.
- [16] S. Beetham, R. O. Fox and J. Capecehatro. Sparse identification of multiphase turbulence closures for coupled fluid-particle flows. *J. Fluid Mech.* 914 (2021), A11.

- [17] R. Beetstra, M. A. van der Hoef and J. Kuipers. Drag force of intermediate Reynolds number flow past mono- and bidisperse arrays of spheres. *AIChE J.* 53 (2) (2007), 489–501.
- [18] S. Boggs. *Principles of Sedimentology and Stratigraphy*. Pearson, London. (2012).
- [19] S. Bogner, S. Mohanty and U. Rde. Drag correlation for dilute and moderately dense fluid–particle systems using the lattice Boltzmann method. *Int. J. Multiphase Flow* 68 (2015), 71–79.
- [20] M. Breuer. Large–eddy simulation of the sub–critical flow past a circular cylinder: Numerical and modeling aspects. *Int. J. Numer. Meth. Fluids* 28 (9) (1998), 1281–1302.
- [21] M. Breuer. A challenging test case for large–eddy simulation: High Reynolds number circular cylinder flow. *Int. J. Heat Fluid Flow* 21 (5) (2000), 648–654.
- [22] M. Breuer. *Direkte Numerische Simulation und Large–Eddy Simulation turbulenter Strmungen auf Hochleistungsrechnern*. Habilitationsschrift, Universitt Erlangen–Nrnberg, Berichte aus der Strmungstechnik. Shaker Verlag, Aachen. (2002).
- [23] M. Breuer and N. Almohammed. Modeling and simulation of particle agglomeration in turbulent flows using a hard–sphere model with deterministic collision detection and enhanced structure models. *Int. J. Multiphase Flow* 73 (2015), 171–206.
- [24] **M. Breuer and A. Khalifa. Revisiting and improving models for the breakup of compact dry powder agglomerates in turbulent flows within Eulerian–Lagrangian simulations. *Powder Technology* 348 (2019), 105–125. DOI: [10.1016/j.powtec.2019.03.009](https://doi.org/10.1016/j.powtec.2019.03.009).**
- [25] **M. Breuer and A. Khalifa. Refinement of breakup models for compact powder agglomerates exposed to turbulent flows considering relevant time scales. *Computers & Fluids* 194 (2019), 104315. DOI: [10.1016/j.compfluid.2019.104315](https://doi.org/10.1016/j.compfluid.2019.104315).**
- [26] **M. Breuer and A. Khalifa. Breakup of agglomerates in turbulent flows: An Euler-Lagrange LES study. In: *Twelfth Int. ERCOFTAC Workshop on DNS and LES: DLES-12, Madrid, Spain, June 5–7, 2019, ERCOFTAC Series, Direct and Large-Eddy Simulation XII, ISBN 978-3-030-42821-1. Vol. 27. Springer Nature Switzerland AG. (2020), 127–133. DOI: [10.1007/978-3-030-42822-8_17](https://doi.org/10.1007/978-3-030-42822-8_17).***
- [27] S. L. Brunton, B. R. Noack and P. Koumoutsakos. Machine learning for fluid mechanics. *Annual Review Fluid Mech.* 52 (2020), 477–508.
- [28] S. L. Brunton, J. L. Proctor and J. N. Kutz. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences* 113 (15) (2016), 3932–3937.
- [29] T. M. Burton and J. K. Eaton. Fully resolved simulations of particle–turbulence interaction. *J. Fluid Mech.* 545 (2005), 67–111.
- [30] G. Calvert, A. Hassanpour and M. Ghadiri. Mechanistic analysis and computer simulation of the aerodynamic dispersion of loose aggregates. *Chemical Eng. Research Design* 89 (5) (2011), 519–525.
- [31] J. S. Capecelatro. A mesoscopic formalism for simulating particle-laden flows with applications in energy conversion processes. PhD thesis. Cornell University, New York, USA, (2014).
- [32] S. Chen and S. Q. Li. Collision–induced breakage of agglomerates in homogeneous isotropic turbulence laden with adhesive particles. *J. Fluid Mech.* 902 (2020), A28.
- [33] T. Chen, Z. Hu, Z. Yang and Y. Zhang. A resolved CFD–DEM investigation into the onset of suffusion: Effect of confining pressure and stress anisotropy. *Int. J. Numerical Analytical Methods Geomech.* 47 (2023), 3018–3043.
- [34] C. Chi, G. Janiga and D. Thvenin. On-the-fly artificial neural network for chemical kinetics in direct numerical simulations of premixed combustion. *Combustion Flame* 226 (2021), 467–477.
- [35] C. Chree. The equilibrium of an isotropic elastic solid ellipsoid under the action of normal surface forces of the second degree, and bodily forces derived from a potential of the second degree. *Quarterly J. Pure Applied Mathematics* 27 (1895), 338–353.
- [36] J. Chun and D. Koch. Coagulation of monodisperse aerosol particles by isotropic turbulence. *Phys. Fluids* 17 (2) (2005), 027102.
- [37] M. Cranmer, A. Sanchez Gonzalez, P. Battaglia, R. Xu, K. Cranmer, D. Spergel and S. Ho. Discovering symbolic models from deep learning with inductive biases. *Advances in Neural Information Processing Systems* 33 (2020), 17429–17442.
- [38] Y. Cui and M. Sommerfeld. Application of lattice-Boltzmann method for analysing detachment of micron-sized particles from carrier particles in turbulent flows. *Flow Turbulence Combustion* 100 (1) (2018), 271–297.
- [39] M. Dietzel, M. Ernst and M. Sommerfeld. Application of the lattice–Boltzmann method for particle-laden flows: Point–particles and fully resolved particles. *Flow Turbulence Combustion* 97 (2) (2016), 539–570.
- [40] M. Dietzel and M. Sommerfeld. Numerical calculation of flow resistance for agglomerates with different morphology by the lattice–Boltzmann method. *Powder Technology* 250 (2013), 122–137.
- [41] F. F. Dizaji, J. S. Marshall and J. R. Grant. Collision and breakup of fractal particle agglomerates in a shear flow. *J. Fluid Mech.* 862 (2019), 592–623.
- [42] K. Duraisamy, G. Iaccarino and H. Xiao. Turbulence modeling in the age of data. *Annu. Rev. Fluid Mech.* 51 (2019), 357–377.
- [43] M. L. Eggersdorfer, D. Kadau, H. J. Herrmann and S. E. Pratsinis. Fragmentation and restructuring of soft-agglomerates under shear. *J. Colloid Interface Science* 342 (2) (2010), 261–268.
- [44] S. Elghobashi and T. Abou-Arab. A two–equation turbulence model for two–phase flows. *Phys. Fluids* 26 (4) (1983), 931–938.

- [45] H. Elmeistikawy, J. Reuter, F. Evrard, S. Mostaghim and B. van Wachem. Deterministic drag modelling for spherical particles in Stokes regime using data-driven approaches. *Int. J. Multiphase Flow* 178 (2024), 104880.
- [46] F. Evrard, F. Denner and B. van Wachem. Euler–Lagrange modelling of dilute particle–laden flows with arbitrary particle–size to mesh–spacing ratio. *J. Comput. Phys.: X* 8 (2020), 100078.
- [47] M. Fanelli, D. L. Feke and I. Manas-Zloczower. Prediction of the dispersion of particle clusters in the nano-scale–Part I: Steady shearing responses. *Chem. Eng. Sci.* 61 (2) (2006), 473–488.
- [48] M. Fanelli, D. L. Feke and I. Manas-Zloczower. Prediction of the dispersion of particle clusters in the nano-scale–Part II, Unsteady shearing responses. *Chem. Eng. Sci.* 61 (15) (2006), 4944–4956.
- [49] C. Fernandes, S. Faroughi, O. Carneiro, J. M. Nóbrega and G. McKinley. Fully-resolved simulations of particle-laden viscoelastic fluids using an immersed boundary method. *J. Non-Newtonian Fluid Mech.* 266 (2019), 80–94.
- [50] G. Frungieri, M. U. Bäbler, L. Biferale and A. S. Lanotte. Heavy and light inertial particle aggregates in homogeneous isotropic turbulence: A study on breakup and stress statistics. *Computers & Fluids* 263 (2023), 105944.
- [51] G. Frungieri, G. Boccardo, A. Buffo, D. Marchisio, H. A. Karimi-Varzaneh and M. Vanni. A CFD-DEM approach to study the breakup of fractal agglomerates in an internal mixer. *The Canadian J. of Chem. Eng.* 98 (9) (2020), 1880–1892.
- [52] G. Frungieri and M. Vanni. Aggregation and breakup of colloidal particle aggregates in shear flow: A combined Monte Carlo-Stokesian dynamics approach. *Powder Technology* 388 (2021), 357–370.
- [53] F. Gong, H. Huang, T. Babadagli and H. Li. A resolved CFD-DEM coupling method to simulate proppant transport in narrow rough fractures. *Powder Technology* 428 (2023), 118778.
- [54] J. Gregory. Fundamentals of flocculation. *Critical Reviews in Environmental Control* 19 (3) (1989), 185–230.
- [55] E. Guesnet, R. Dendievel, D. Jauffrès, C. Martin and B. Yrieix. A growth model for the generation of particle aggregates with tunable fractal dimension. *Physica A: Statistical Mechanics and its Applications* 513 (2019), 63–73.
- [56] A. Hager, C. Kloss, S. Pirker and C. Goniva. Parallel resolved open source CFD-DEM: Method, validation and application. *J. Computational Multiphase Flows* 6 (1) (2014), 13–27.
- [57] M. Hansinger, Y. Ge and M. Pfitzner. Deep residual networks for flamelet/progress variable tabulation with application to a piloted flame with inhomogeneous inlet. *Combustion Science Technology* (2020), 1–27.
- [58] S. Harada, R. Tanaka, H. Nogami and M. Sawada. Dependence of fragmentation behavior of colloidal aggregates on their fractal structure. *J. Colloid Interface Science* 301 (1) (2006), 123–129.
- [59] R. Hassanian, Á. Helgadóttir, F. Gharibi, A. Beck and M. Riedel. Data-driven deep learning models in particle-laden turbulent flow. *Phys. Fluids* 37 (2) (2025).
- [60] K. Higashitani, K. Iimura and H. Sanda. Simulation of deformation and breakup of large aggregates in flows of viscous fluids. *Chem. Eng. Sci.* 56 (9) (2001), 2927–2938.
- [61] C. A. Ho and M. Sommerfeld. Modelling of micro–particle agglomeration in turbulent flows. *Chem. Eng. Sci.* 57 (2002), 3073–3084.
- [62] A. Hölzer and M. Sommerfeld. Lattice Boltzmann simulations to determine drag, lift and torque acting on non-spherical particles. *Computers & Fluids* 38 (3) (2009), 572–589.
- [63] F. Hoppe and M. Breuer. A deterministic and viable coalescence model for Euler–Lagrange simulations of turbulent microbubble–laden flows. *Int. J. Multiphase Flow* 99 (2018), 213–230.
- [64] A. S. Jebakumar, K. N. Premnath, V. Magi and J. Abraham. Fully–resolved direct numerical simulations of particle motion in a turbulent channel flow with the lattice–Boltzmann method. *Computers & Fluids* 179 (2019), 238–247.
- [65] T. Kempe and J. Fröhlich. An improved immersed boundary method with direct forcing for the simulation of particle laden flows. *J. Comput. Phys.* 231 (9) (2012), 3663–3684.
- [66] T. Kempe and J. Fröhlich. Collision modelling for the interface–resolved simulation of spherical particles in viscous fluids. *J. Fluid Mech.* 709 (2012), 445–489.
- [67] **A. Khalifa. Modeling and simulation of the breakage of cohesive particle agglomerates in turbulent wall-bounded flows. PhD thesis. Department of Fluid Mechanics, Helmut-Schmidt-University, Hamburg, Germany, (2022). DOI: 10.24405/14518.**
- [68] **A. Khalifa and M. Breuer. Data–driven model for the breakage of dry monodisperse agglomerates by wall impact applicable for multiphase flow simulations. Powder Technology 376 (2020), 241–253. DOI: 10.1016/j.powtec.2020.07.089.**
- [69] **A. Khalifa and M. Breuer. An efficient model for the breakage of agglomerates by wall impact applied to Euler-Lagrange LES predictions. Int. J. Multiphase Flow 142 (2021), 103625. DOI: 10.1016/j.ijmultiphaseflow.2021.103625.**
- [70] **A. Khalifa and M. Breuer. Data-driven ANN approach for binary agglomerate collisions including breakage and agglomeration. Chemical Engineering Research Design 95 (2023), 14–27. DOI: 10.1016/j.cherd.2023.05.040.**
- [71] A. Khalifa and M. Breuer. Euler–Lagrange LES predictions of a powder disperser including a multiscale wall-impact breakage model. In: *13th Int. ERCOFTAC Workshop on DNS and LES: DLES-13, Udine, Italy, Oct. 26–29, 2022, ERCOFTAC Series, ISBN 978–3–031–47027–1. Vol. 31. Springer Nature Switzerland. (2024), 89–94.*
- [72] **A. Khalifa and M. Breuer. Evaluation of an efficient data-driven ANN model to predict agglomerate collisions within Euler–Lagrange simulations. Computers & Fluids 269 (2024), 106119. DOI: https://doi.org/10.1016/j.compfluid.2023.106119.**

- [73] **A. Khalifa, M. Breuer and J. Gollwitzer. Neural-network based approach for modeling wall-impact breakage of agglomerates in particle-laden flows applied in Euler-Lagrange LES. *Int. J. Heat Fluid Flow* 94 (2022), 108897. DOI: [10.1016/j.ijheatfluidflow.2021.108897](https://doi.org/10.1016/j.ijheatfluidflow.2021.108897).**
- [74] **A. Khalifa, J. Gollwitzer and M. Breuer. LES of particle-laden flow in sharp pipe bends with data-driven predictions of agglomerate breakage by wall impacts. *Fluids* 6 (12) (2021), 424. DOI: [10.3390/fluids6120424](https://doi.org/10.3390/fluids6120424).**
- [75] A. Khalifa and M. Breuer. *Investigation of cohesive particle deagglomeration in homogeneous isotropic turbulence using particle-resolved DNS*. Under second-round review at Phys. Fluids. DOI: [10.48550/arXiv.2505.18707](https://doi.org/10.48550/arXiv.2505.18707).
- [76] C. Kloss, C. Goniva, A. Hager, S. Amberger and S. Pirker. Models, algorithms and validation for opensource DEM and CFD-DEM. *Progress Comput. Fluid Dyn. An Int. J.* 12 (2-3) (2012), 140–152.
- [77] A. R. Kobaltz, S. Lovett, N. Nikiforakis and W. D. Henshaw. Direct numerical simulation of particulate flows with an overset grid method. *J. Comp. Physics* 343 (2017), 414–431.
- [78] P. Kosinski and A. C. Hoffmann. Extended hard-sphere model and collisions of cohesive particles. *Phys. Rev. E* 84 (3) (2011), 031303.
- [79] M. J. Kunz. Effect of large dams in the Zambezi River basin: Changes in sediment, carbon and nutrient fluxes. PhD thesis. ETH Zürich, Switzerland, (2011).
- [80] R. Kurose and S. Komori. Drag and lift forces on a rotating sphere in a linear shear flow. *J. Fluid Mech.* 384 (1999), 183–206.
- [81] K. A. Kusters. The influence of turbulence on aggregation of small particles in agitated vessels. PhD thesis. Technische Universiteit Eindhoven, The Netherlands, (1991).
- [82] P. W. Longest, Y.-J. Son, L. Holbrook and M. Hindle. Aerodynamic factors responsible for the deaggregation of carrier-free drug powders to form micrometer and submicrometer aerosols. *Pharmaceutical Research* 30 (6) (2013), 1608–1627.
- [83] T. Ma, T. Ziegenhein, D. Lucas, E. Krepper and J. Fröhlich. Euler-Euler large eddy simulations for dispersed turbulent bubbly flows. *Int. J. Heat Fluid Flow* 56 (2015), 51–59.
- [84] C. Marchioli and A. Soldati. Turbulent breakage of ductile aggregates. *Physical Review E* 91 (5) (2015), 053003.
- [85] D. L. Marchisio, J. T. Pikturna, R. O. Fox, R. D. Vigil and A. A. Barresi. Quadrature method of moments for population-balance equations. *AIChE J.* 49 (5) (2003), 1266–1276.
- [86] E. Masi, O. Simonin, E. Riber, P. Sierra and L. Y. Gicquel. Development of an algebraic closure-based moment method for unsteady Eulerian simulations of particle-laden turbulent flows in very dilute regime. *Int. J. Multiphase Flow* 58 (2014), 257–278.
- [87] F. J. Montáns, F. Chinesta, R. Gómez-Bombarelli and J. N. Kutz. Data-driven modeling and learning in science and engineering. *Comptes Rendus Mécanique* 347 (11) (2019), 845–855.
- [88] M. Moreau, O. Simonin and B. Bédard. Development of gas-particle Euler-Euler LES approach: A-priori analysis of particle sub-grid models in homogeneous isotropic turbulence. *Flow Turbulence Combustion* 84 (2) (2010), 295.
- [89] K. Mühle. Floc stability in laminar and turbulent floc. In: *Coagulation and Flocculation*. Ed. by B. Dobia. Decker, New York. (1993), 355–390.
- [90] T. Neeße, K. Mühle and A. Ivanauskas. Zur physikalisch begründeten Modellierung des Flockungsprozesses, Teil V: Flockenaufbruch und Flockenerosion in turbulenter Strömung. *Chem.-Ing.-Techn.* 39 (1987), 292–295.
- [91] P. Neumann, J. A. Duffek, J. Kleinschmidt, W. G. Leinen, M. Breuer, S. Schmidt-Lauff, A. Fink, M. Mayr, M. Firmbach, A. Popp and A. Auweter. hpc.bw: A Supercomputer with Competence Platform for the Universities of the Federal Armed Forces. In: *dtec-Beiträge der Helmut-Schmidt-Universität Hamburg: Forschungsaktivitäten im Zentrum für Digitalisierungs- und Technologieforschung der Bundeswehr dtec.bw*. Vol. 1. (2022), 305–310.
- [92] D. Ramkrishna. *Population Balances: Theory and Applications to Particulate Systems in Engineering*. Elsevier. (2000).
- [93] H. Rumpf. The strength of granules and agglomerates. *Agglomeration – Proc. of the First Int. Symp. on Agglomeration*, (ed. W.A. Knepper), Interscience, New York (1962), 379–418.
- [94] H. Rumpf and J. Raasch. Desagglomeration in Strömungen. In: *1. Europ. Symp. Zerkleinern 10.-13.4.1962*. Verlag Chemie, Weinheim und VDI-Verlag, Düsseldorf. (1962), 151–159.
- [95] D. Saha. Experimental analysis of aggregate breakup in flows observed by three dimensional particle tracking velocimetry. PhD thesis. ETH Zürich, Switzerland, (2013).
- [96] D. Saha, M. U. Bäbler, M. Holzner, M. Soos, B. Lüthi, A. Liberzon and W. Kinzelbach. Breakup of finite-size colloidal aggregates in turbulent flow investigated by three-dimensional (3D) particle tracking velocimetry. *Langmuir* 32 (1) (2015), 55–65.
- [97] A. Saxena, J.-S. Kroll-Rabotin and R. S. Sanders. Numerical investigation of the respective roles of cohesive and hydrodynamic forces in aggregate restructuring under shear flow. *J. Colloid Interface Science* 608 (2022), 355–365.
- [98] A. Saxena, J.-S. Kroll-Rabotin and R. S. Sanders. Role of flow inertia in aggregate restructuring and breakage at finite Reynolds Numbers. *Langmuir* 39 (29) (2023), 10066–10078.
- [99] A. Saxena, J.-S. Kroll-Rabotin and R. S. Sanders. Exposure of fractal aggregates to accelerating flows at finite Reynolds numbers. *Int. J. Multiphase Flow* 182 (2025), 105018.
- [100] K. C. J. Schutte, L. M. Portela, A. Twerda and R. A. W. M. Henkes. Formation and break-up of rigid agglomerates in turbulent channel and pipe flows. *J. Fluid Mech.* 857 (2018), 539–561.

- [101] K. C. J. Schutte, L. M. Portela, A. Twerda and R. A. W. M. Henkes. Hydrodynamic perspective on asphaltene agglomeration and deposition. *Energy & Fuels* 29 (5) (2015), 2754–2767.
- [102] A. Schwarz, P. Kopper, J. Keim, H. Sommerfeld, C. Koch and A. Beck. A neural network based framework to model particle rebound and fracture. *Wear* 508 (2022), 204476.
- [103] O. L. Sgrott. Influence of Inter-Particle Interactions on the Performance of Cyclone Separators. PhD thesis. Faculty of Process- and Systems Engineering, Otto-von-Guericke-University Magdeburg, Magdeburg, Germany, (2019).
- [104] O. L. Sgrott and M. Sommerfeld. Influence of inter-particle collisions and agglomeration on cyclone performance and collection efficiency. *The Canadian J. of Chem. Eng.* 97 (2) (2019), 511–522.
- [105] Y. Shao, G. McTainsh, J. Leys and M. Raupach. Efficiencies of sediment samplers for wind erosion measurement. *Soil Research* 31 (4) (1993), 519–532.
- [106] Z. Shen, G. Wang, D. Huang and F. Jin. A resolved CFD-DEM coupling model for modeling two-phase fluids interaction with irregularly shaped particles. *J. Comp. Phys.* 448 (2022), 110695.
- [107] A. P. Singh, S. Medida and K. Duraisamy. Machine-learning-augmented predictive modeling of turbulent separated flows over airfoils. *AIAA J.* 55 (7) (2017), 2215–2227.
- [108] R. C. Sonntag and W. B. Russel. Structure and breakup of flocs subjected to fluid stresses: II. Theory. *J. Colloid Interface Science* 115 (2) (1987), 378–389.
- [109] E. Strohmaier, J. Dongarra, H. Simon and M. Meuer. *TOP500 – June 2002*. Tech. rep. 2022.
- [110] D. G. Thomas. Transport characteristics of suspensions VII. Relation of hindered-settling floc characteristics to rheological parameters. *AIChE J.* 9 (3) (1963), 310–316.
- [111] D. G. Thomas. Turbulent disruption of flocs in small particle size suspensions. *AIChE J.* 10 (4) (1964), 517–523.
- [112] D. T. Tomi and D. F. Bagster. The behavior of aggregates in stirred vessels. *Transactions of the Institution of Chemical Engineers* 56 (1) (1978), 9–18.
- [113] Z. Tong, B. Zheng, R. Yang, A. Yu and H.-K. Chan. CFD–DEM investigation of the dispersion mechanisms in commercial dry powder inhalers. *Powder Technology* 240 (2013), 19–24.
- [114] M. Uhlmann. An immersed boundary method with direct forcing for the simulation of particulate flows. *J. Comput. Phys.* 209 (2) (2005), 448–476.
- [115] M. Uhlmann. Interface-resolved direct numerical simulation of vertical particulate channel flow in the turbulent regime. *Phys. Fluids* 20 (5) (2008), 053305.
- [116] B. van Wachem, K. Thalberg, D. Nguyen, L. M. de Juan, J. Remmelgas and I. Niklasson-Bjorn. Analysis, modelling and simulation of the fragmentation of agglomerates. *Chem. Eng. Sci.* 227 (2020), 115944.
- [117] B. van Wachem, K. Thalberg, J. Remmelgas and I. Niklasson-Björn. Simulation of dry powder inhalers: Combining micro-scale, meso-scale and macro-scale modeling. *AIChE J.* 63 (2) (2017), 501–516.
- [118] B. Vowinkel, J. Withers, P. L.-Fegiz and E. Meiburg. Settling of cohesive sediment: Particle-resolved simulations. *J. Fluid Mech.* 858 (2019), 5–44.
- [119] B. Vowinkel, E. Biegert, P. Luzzatto-Fegiz and E. Meiburg. Consolidation of freshly deposited cohesive and non-cohesive sediment: Particle-resolved simulations. *Phys. Rev. Fluids* 4 (7) (2019), 074305.
- [120] A. Vreman. Particle-resolved direct numerical simulation of homogeneous isotropic turbulence modified by small fixed spheres. *J. Fluid Mech.* 796 (2016), 40–85.
- [121] C. Weiler. Generierung leicht dispergierbarer Inhalationspulver mittels Sprühtrocknung. PhD thesis. Johannes Gutenberg-Universität Mainz, Mainz, Germany, (2008).
- [122] C. Weiler, M. Wolkenhauer, M. Trunk and P. Langguth. New model describing the total dispersion of dry powder agglomerates. *Powder Technology* 203 (2) (2010), 248–253.
- [123] J. Williams, U. Wolfram and A. Ozel. Neural stochastic differential equations for particle dispersion in large-eddy simulations of homogeneous isotropic turbulence. *Phys. Fluids* 34 (11) (2022), 113315.
- [124] Y. Y. Tang, E. F. Peters, J. H. Kuipers, S. Sebastian Kriebitzsch and M. Martin van der Hoef. A new drag correlation from fully resolved simulations of flow past monodisperse static arrays of spheres. *AIChE J.* 61 (2) (2015), 688–698.
- [125] R. Y. Yang, A. B. Yu, S. K. Choi, M. S. Coates and H. K. Chan. Agglomeration of fine particles subjected to centripetal compaction. *Powder Technology* 184 (1) (2008), 122–129.
- [126] Y. Yao and J. Capecehatro. Deagglomeration of cohesive particles by turbulence. *J. Fluid Mech.* 911 (2021).
- [127] A. Yeung, A. Gibbs and R. Pelton. Effect of shear on the strength of polymer-induced flocs. *J. Colloid Interface Science* 196 (1) (1997), 113–115.
- [128] X. Yu, M. Yu, S. Balachandar and A. Manning. Shielding effects of neighbor particles on flocculation dynamics of cohesive sediment. *Phys. Fluids* 37 (3) (2025), 033369.
- [129] A. Zacccone, M. Soos, M. Lattuada, H. Wu, M. U. Bäbler and M. Morbidelli. Breakup of dense colloidal aggregates under hydrodynamic stresses. *Physical Review E* 79 (6) (2009), 061401.
- [130] K. Zhao, F. Pomes, B. Vowinkel, T.-J. Hsu, B. Bai and E. Meiburg. Flocculation of suspended cohesive particles in homogeneous isotropic turbulence. *J. Fluid Mech.* 921 (2021), A17.
- [131] Z. Zheng, S. S. Y. Leung and R. Gupta. Flow and particle modelling of dry powder inhalers: Methodologies, recent development and emerging applications. *Pharmaceutics* 13 (2) (2021), 189.
- [132] L. Zhou. Advances in studies on two-phase turbulence in dispersed multiphase flows. *Int. J. Multiphase Flow* 36 (2) (2010), 100–108.

4 Supplementary information on the research context

4.1 Ethical and/or legal aspects of the project

4.1.1 General ethical aspects

No statement by an ethics committee is required.

4.1.2 Descriptions of proposed investigations involving humans, human materials or identifiable data

Not applicable

4.1.3 Descriptions of proposed investigations involving experiments on animals

Not applicable

4.1.4 Descriptions of projects involving genetic resources (or associated traditional knowledge) from a foreign country

Not applicable

4.1.5 Explanations regarding any possible safety-related aspects

4.1.5.1 “Dual Use Research of Concern”; foreign trade law

Not applicable

4.1.5.2 Risks in international cooperation

Not applicable

4.1.6 Considerations on aspects of ecological sustainability in the planning and implementation of the project

In planning and implementing the proposed project, we have carefully considered ecological sustainability, particularly in terms of travel and computational resource usage.

Travel Activities: The project includes a travel budget intended primarily for essential scientific exchange and networking, especially for early-career researchers. Whenever possible, we will prioritize digital communication (e.g., virtual meetings and workshops) to minimize travel-related emissions. In cases where physical travel is necessary, we will aim to combine multiple goals per trip, choose direct flights when available, and explore alternative lower-emission transport modes such as a train journey. Hybrid or fully virtual formats will be preferred for project-related events such as workshops and symposia to reduce travel needs.

Computational Resources: Given the computationally intensive nature of DNS-DEM simulations, we will implement several measures to reduce environmental impact:

- All simulations will be hypothesis-driven and planned to avoid redundant computations.

- We will monitor and document our computing resource usage (e.g., core hours, CPU time,) and estimate the associated CO₂ emissions for increased awareness (now available in SLURM). Our compute cluster HSUper was recently listed in the Green-Top500 list due to high energy efficiency achieved by cold water cooling.
- Reusability of simulation data will be ensured through thorough documentation, centralized storage, and potential public sharing, thus reducing the need for future redundant simulations.

Overall, our approach is to achieve scientific excellence while maintaining a conscious and responsible use of resources throughout the project lifecycle.

4.2 Employment Status Information

Name:	Dr.-Ing. Ali Khalifa
Official position:	Scientific co-worker (PostDoc)
Employment status:	Fixed-term contract presently until 31.12.2026
Institution:	Helmut-Schmidt-University Hamburg (HSU)

Name:	Univ.-Prof. Dr.-Ing. habil. Michael Breuer
Official position:	Permanent professor, head of the department
Institution:	Helmut-Schmidt-University Hamburg (HSU)

4.3 First-time proposal data

Dr.-Ing. Khalifa, Ali

Dr.-Ing. A. Khalifa is a postdoctoral researcher at the department. He has a demonstrated experience in the multiscale modeling and simulation of cohesive particle-laden flows using data-driven approaches. He will supervise the project together with Prof. Breuer.

Dr.-Ing. Khalifa finished his PhD in September 2022 at HSU Hamburg with the highest grade, i.e. *summa cum laude*. He then successfully applied for a one-year funding at the *Interne Forschungsförderung (IFF)* of HSU. The IFF funding was installed at HSU a few years ago to give young and promising scientists the chance to conscientiously prepare and submit their own DFG application. After a competitive selection process with a written application and an oral presentation of the project idea, the proposal of Dr.-Ing. Khalifa was selected from a variety of suggested projects for a start-up funding of 1 year (July 1, 2023 to June 30, 2024). This funding is linked to the explicit requirement to submit a DFG application during the course of the project. In spring 2024 Dr.-Ing. Khalifa received the "Wissenschaftspreis der Gesellschaft der Freunde und Förderer der HSU Hamburg" for his excellent PhD thesis.

Prior to this proposal, Dr.-Ing. Khalifa submitted a related proposal to the DFG, along with a revised version, which was ultimately not recommended for funding. The main criticism of the original submission was the use of neural networks for developing the deagglomeration model, which was viewed as lacking physical interpretability. In the revised version with new reviewers, although this issue was fully addressed, the proposal was considered to place too much emphasis on modeling and not enough on the physical analysis of the deagglomeration process.

In the new proposal, both points of criticism have been carefully addressed. First, we prioritize the use of symbolic regression and plan to evaluate various techniques for extracting models from particle-resolved simulation data. This approach ensures that the resulting model is physically interpretable, as demonstrated in the attached preliminary study. Furthermore, the proposal places greater emphasis on the physical mechanisms underlying deagglomeration, ensuring a balanced integration of modeling and physical analysis.

4.4 Composition of the Project Group

Univ.-Prof. Dr.-Ing. habil. M. Breuer is the head of the Department of Fluid Mechanics at the HSU Hamburg. Together with Dr. Khalifa he supervises the project and support the PhD candidate during the project.

4.5 Researchers in Germany with whom you have agreed to cooperate on this project

None

4.6 Researchers abroad with whom you have agreed to cooperate on this project

None

4.7 Researchers with whom you have collaborated scientifically within the past three years

Collaborations by Prof. Dr.-Ing. habil. M. Breuer:

- Prof. Dr. K.U. Bletzinger, Lehrstuhl für Statik, Technische Universität München
- Dr. Ann-Kathrin Goldbach, Lehrstuhl für Statik, Technische Universität München
- Dr. Andreas Apostolatos, MathWorks
- Prof. Dr. D. S. Henningson and Prof. Dr. A. Hanifi, KTH Royal Institute of Technology, Dept. of Eng. Mechanics, Stockholm, Sweden
- Prof. Dr. P. Schaffarczyk, Kiel University of Applied Sciences, Kiel
- Prof. Dr. P. Neumann, High-Performance Computing, Helmut-Schmidt Universität Hamburg
- Prof. Dr. O. Niggemann, Informatik in Maschinenbau, Helmut-Schmidt Universität Hamburg
- Prof. Dr. W. E. Weber, Statik und Dynamik, Helmut-Schmidt Universität Hamburg
- Prof. Dr. J. Küpper, Center for Free-Electron Laser Science, DESY, Hamburg
- Prof. Dr. G. Biswas, Department of Mech. Eng., Indian Institute of Technology Kanpur, India
- Dr. B. Ünsal, TÜBİTAK UME (National Metrology Institute of Turkey), Kocaeli, Turkey

4.8 Project-relevant cooperation with commercial enterprises

None

4.9 Project-relevant participation in commercial enterprises

None

4.10 Scientific equipment

For the development of the methods and tools as well as for the pre- and post-processing two different in-house Linux clusters with 960/512 cores are available at the department. Furthermore, the university has recently installed the HPC-system HSUPER[†] which can be used. It consists of 571 compute nodes each equipped with 256 GB RAM and 2 Intel Icelake sockets. Each socket features a Intel(R) Xeon (R) Platinum 8360Y processor with 36 cores, yielding a total of 72 cores per node. All nodes are connected by a non-blocking InfiniBand HDR100 fabric network. HSUPER has entered the TOP500 list of the fastest supercomputers in the world in June 2022, ranked 339 with a Linpack performance of 2.13 PFLOPS [109]. It is further ranked 70 in the Green500 list, which measures the computational performance per consumed Watt [91]. In case these resources are still not sufficient, an application will be sent to one of the new National High Performance Computing Centers (NHR). HSU Hamburg is presently also extending their hardware equipment regarding CPUs and GPUs. The latter will be advantageous for the machine-learning algorithms to be applied in this project.

4.11 Other submissions

None

4.12 Other information

None

[†]<https://www.hsu-hh.de/hpc/en/hsuper>

5 Requested modules / funds

5.1 Basic module

5.1.1 Funding for staff

- 1 Doctoral researcher (TV-L E 13, full time for 36 months)
- 1 Student assistant 10 h/week for 36 months
 - Description: The participation of a student assistant is strongly required since the simulation tasks and the data analysis tasks within this project are quite extensive. Here the researcher should be supported by a student assistant. He/she can also assist the testing and validation phase of the methods and algorithms developed within the project.
 - Explanation: The doctoral researcher will be supervised in close collaboration between Dr. Khalifa and Prof. Breuer. However, Dr. Khalifa will assume sole personnel responsibility for the student employee who supports the academic employee. Since he has already worked on the topic as a part of his postdoctoral studies and is thus deeply rooted in the subject matter, he will assume full supervision of the student assistant.

5.1.2 Direct project costs

5.1.2.1 Equipment up to 10,000 €, software and consumables

None

5.1.2.2 Travel expenses

Funding for travel is necessary to present the results of the project at several national (e.g., *GAMM*) and international conferences (e.g., *Int. Conference on Multiphase Flow, DLES*), at least once per year.

3 000 € / year

5.1.2.3 Visiting researchers (excluding Mercator Fellows)

None

5.1.2.4 Expenses for laboratory animals

None

5.1.2.5 Other costs

None

5.1.2.6 Project-related publication expenses

In order to publish the outcome of the project in international journals, additional funding is required:

750 € / year

The authors started to publish with open access and observed a very positive impact on the knowledge dispersion in the short term compared to subscription-only work.

5.1.3 Instrumentation

5.1.3.1 Equipment exceeding 10,000 €

None

5.1.3.1 Major instrumentation exceeding 50,000 €

None

5.2 Module Temporary Position for Principal Investigator

None

5.3 Module Replacement Funding

None

5.4 Module Temporary Clinician Substitute

None

5.5 Module Mercator Fellows

None

5.6 Module Workshop Funding

None

5.7 Module Public Relations Funding

None

5.8 Module Standard Allowance for Equity and Diversity

None